Interpreting the Exponential Decay Fit $\left(y = a \cdot e^{-bx}\right)$

When you say you have:

1. A **cluster size distribution** from size 1 up to 10,
2. **Cluster size 1** (i.e., single-pellet events) being the most frequent,
3. Then sizes 2, 3, 4, 5 with somewhat closer counts,
4. A noticeable drop (or "gap") around size 6 and higher,
5. A good **exponential decay fit** with parameters $a = 139$ and $b = 0.28$, and an $R^2 = 0.96$,

you are essentially describing how the **frequency** (or count) of clusters falls off as cluster size increases.

1. What the Parameters Mean

An exponential function

$$y \;=\; a\, e^{-bx}$$

often fits count data when there is a rapid drop-off with increasing x. In this context:

- $x =$ cluster size (1, 2, 3, ...).
- $y =$ frequency (how many clusters) of each size x.

So, for each cluster size x,

$$y \;\approx\; a \,\cdot\, e^{-bx}$$

is your fitted count.

- $\boxed{a = 139}$: This is the **scale factor**, basically saying that at $x = 0$ (hypothetically), y would be 139. Of course, you do not actually have a "size 0" cluster, but a sets the overall height of the distribution at the smallest cluster sizes.
- $\boxed{b = 0.28}$: This is the **rate** at which the exponential decays. A smaller b (closer to 0) would decay more slowly, and a larger b would decay more quickly. Having $b = 0.28$ means that each increment in cluster size multiplies the expected frequency by roughly

  $$e^{-0.28} \;\approx\; 0.76.$$

  In other words, you lose about **24%** of your count for each increase by 1 in cluster size, according to the fitted curve.

A **high** $R^2 = 0.96$ implies the exponential model explains about 96% of the variance in how often each cluster size appears.

2. How This Relates to Your Observations

1. **Single-Pellet Events Are Very Common**
   Because when $x = 1$, the exponential fit is

   $$y = a \cdot e^{-b \cdot 1}.$$

   Substituting $a = 139$ and $b = 0.28$ gives

   $$139 \times e^{-0.28} \approx 139 \times 0.76 \approx 106.$$

   That is fairly high compared to the counts for larger cluster sizes (where you keep multiplying by additional factors of $e^{-0.28}$).

2. **Clusters of Size 2–5 Are in a Moderate Range**
   If you keep incrementing x, the model's frequency gradually (or somewhat quickly) decays. So from size 2 up to 5, you would see smaller but still noticeable counts.

3. **"Gap" or Lower Frequencies for 6+**
   Once you hit 6 and beyond, the exponential curve is low enough that you see a clear drop-off. That may align with your empirical observation that cluster sizes above 5 are uncommon or "gapped."

Thus, the **exponential decay** with $b = 0.28$ says, "As cluster size grows by 1, the expected number of clusters is multiplied by about 0.76." Starting with the highest frequency near size 1 (and still decent counts at sizes 2–5) fits the shape you described.

3. Does This Mean "Single Pellet Events Stand Out"?

Because the exponential fit is so good ($R^2 = 0.96$), it means the **entire distribution** roughly follows the same rule: each additional pellet in the cluster leads to a consistent fractional drop in frequency.

However, **in many feeding studies**, it is indeed common to see:

- A very large number of **single-pellet** events (often called "isolated snacking" or "small events"),
- Followed by a tail of multi-pellet clusters (2, 3, 4, ...).

Whether you treat size-1 clusters as "snacks" is partly a conceptual or behavioral decision. The exponential model merely tells you that the frequency keeps decreasing in a predictable fashion. If **size-1** is disproportionately high relative to the rest of the curve, you might see:

- A good fit for sizes 2–10, but an *even higher* actual count at size 1.
- A reason to classify "single pellets" separately as their own behavior or "snack-like" event.

If, on the other hand, the exponential fit *already* accommodates size 1 nicely (the model predictions for size 1 are close to the actual count), then "size 1 stands out" only in the sense that it is the largest single bar of the distribution but still on the same exponential trajectory.

In your case, the best approach depends on the exact data. If you see that your real data's count at **size = 1** is *far above* what the exponential curve would predict, that strongly suggests single-pellet events are a special category. If the model's predicted count for size 1 is *about the same* as your observed count, you could interpret it as "all sizes 1 through 10 lie on the same smooth exponential slope, with no special discontinuity for 1."

4. Conclusion

- **Parameter** $b = 0.28$ indicates the rate at which cluster counts drop as size increases. A 0.28 decay parameter means each additional pellet reduces the frequency to about 76% of the previous cluster size.

- **Yes, single-pellet events can be singled out** if you see that size-1 clusters are extremely common relative to the rest. This often aligns with "snacking," whereas clusters of size 2–5 might be mild "meals," and 6+ might be "large meals."

- Ultimately, **how you categorize** "snacks vs. meals vs. big meals" depends on both the distribution shape (which your exponential is capturing nicely) *and* your behavioral definitions. The exponential fit alone does not force you to treat size 1 events separately; it only tells you how often each cluster size occurs in relation to others. If you conceptually want to say "isolated single pellets are a different phenomenon," that is a **study design choice** and is often justified in feeding studies.